

Gonçalo L. Fernandes

Porto, Portugal

+351 923 137 410 | goncalopesfernandes@gmail.com | [linkedin.com/in/gonzalo-aiml-dev](https://www.linkedin.com/in/gonzalo-aiml-dev)

SUMMARY

AI/ML Developer building **production-grade intelligent systems** on Azure Cloud. I design, train, and deploy AI pipelines that process real enterprise data, specializing in **multi-model financial document processing** with **OCR, NLP**, and **GenAI**, as well as in custom **Computer Vision** and **Deep Learning**, for B2B enterprise clients across European markets. In Computer Vision, I build and train custom **object detection pipelines** for drone-based warehouse inventory auditing, fine-tuned end-to-end on real client data.

SKILLS

AI/ML & Document Processing: Azure DI Studio, Azure AI Foundry, scikit-learn, Custom AI Models, ML Model Training, Document Classification, OCR, IDP, Multi-Model AI, AI Orchestration, Pattern Recognition

Computer Vision & Deep Learning: PyTorch, TensorFlow, Ultralytics YOLO, OpenCV, ONNX, LayoutLMv3, Object Detection, Image Classification, Fine-tuning, Custom Training Pipelines

LLM & GenAI: GPT-4.1, Anthropic Claude API, Meta Llama, Hugging Face, Prompt Engineering, LLM Integration, LLM Cost Optimization, Local LLM Deployment, Semantic Analysis, NLP

Cloud & DevOps: Azure Cloud, Azure Logic Apps, Azure Blob Storage, Azure DevOps, Azure Static Web Apps, CI/CD Pipelines, Cloud Architecture

Development: Python, Angular 19, TypeScript, NumPy, Pandas, Microservices, API Integration, JSON, Real-time Systems

Tools & Platforms: Microsoft Graph API, Azure OpenAI Service, Azure Machine Learning, Git, Redis

WORK EXPERIENCE

COSMOS-OMNI

Mar 2025 – Present

AI/ML Developer - Permanent (Aug 2026 – Present)

Porto Metropolitan Area · Hybrid

Permanent position after one year as Junior Developer, continued ownership of the **AEI (Accounting Engine Interface) document processing ecosystem** (Azure Logic Apps, Azure DI Studio, Azure AI Foundry, GPT-4.1), the **Angular 19 enterprise document portal**, and the drone-based **Computer Vision warehouse auditing pipeline** for fashion retail.

AI/ML Junior Developer (Aug 2025 – Jul 2026)

- Development and deployment of **AI automation systems** for document processing and invoice classification, with support for **~110 supplier VAT numbers** in Azure Logic Apps, real-time processing with **40% reduction in manual errors** and **65% reduction in processing time**.
- **Azure DI Studio** integration, custom model training and classification model retraining in **Azure AI Foundry**, multi-model AI orchestration and LLM integration (**GPT-4.1**) for production environments.
- Frontend migration of an enterprise document management portal from legacy ASP.NET MVC to **Angular 19 SPA**, with **OAuth2 Authorization Code + PKCE** authentication through **Azure API Management**.
- **Computer Vision pipeline** for drone-based warehouse auditing: custom-trained **YOLOv2** detection models (4 classes, **~12,000 annotations**) validated end-to-end on real imagery from a fashion retail client, batch inference with **Azure Blob Storage** integration and WMS submission via Azure API Management.
- **Azure DevOps CI/CD** pipelines and **Azure Static Web Apps** deployment, Blob Storage pipeline management and standardized field taxonomies for API integration.

AI/ML Intern Developer (Mar 2025 – Jul 2025)

- API integration and automated deployment pipelines, **Azure Logic Apps** and **Blob Storage** workflow development, Document AI model training and optimization.
- Process automation and workflow orchestration, real-time systems development and monitoring, scalable architecture design and implementation.

Andreia Ferreira Serviços & Consultoria

Jul 2026 – Present

Process Automation & IT Support (Freelance)

Vila Nova de Famalicão, Portugal · Hybrid

- Independent service provider for an administrative and solicitadoria practice supporting citizens and businesses with social security, tax, registry and documentation processes, **standardised workflow** for translating and normalising foreign tax documentation, reducing turnaround time on cross-border cases.

Worten Portugal

Feb 2025 – Aug 2025

Technical Sales Representative in IT (Part-time)

Porto, Portugal

- Technical consultation and product recommendations based on customer needs at Portugal's leading tech retailer (+230 **stores**, +4,000 **employees**), stock and display management, knowledge updates on technology trends, promotion of additional services (Worten Resolve, warranties, insurance), and customer relationship building with loyalty programs.

Ibersol Group

Aug 2024 – Feb 2025

Customer Service in Hospitality (Part-time)

Vila Nova de Gaia, Portugal

- Promotion of services and products to enhance customer experience at Portugal's leading organized restaurant group, attention to detail in service delivery and presentation, customer loyalty initiatives, and teamwork in a fast-paced environment.

Famous Determination

Jul 2023 – Oct 2023

Electrical Collaborator

Portugal (nationwide)

- Electrical installations in residential and commercial buildings, supporting senior electricians in maintenance and repair work across Portugal, installation of lighting, outlets, and distribution panels with **RTIEBT compliance** (Portuguese electrical safety regulations).

PC & Companhia

Mar 2023 – Jul 2023

IT Technician (Internship)

Porto, Portugal

- OS installation and configuration (Windows, macOS, Linux), maintenance and repair of computers, laptops, and electronic equipment, hardware troubleshooting, diagnosis of hardware/software failures, and data backup/recovery services at a specialized computer repair company.

PROJECTS

All projects developed at COSMOS-OMNI.

SkyVision Auditor - Drone-Based Warehouse Auditing with Computer Vision

Computer Vision | YOLO26 Custom | PyTorch | Ultralytics | Python | Azure Blob Storage | Azure APIM

- End-to-end Computer Vision pipeline automating physical warehouse audits with drone-captured imagery across multi-zone fashion retail operations (~6,000 SKUs, ~50,000 units), with CV processing kept on-premise.
- Custom **YOLO26** detection model (4 classes: box, clothing, footwear, empty_space) with iterative dataset versioning, **filename-anchored location mapping** tying each frame to structured WMS codes - no barcodes, ArUco markers, or 3D scene reconstruction required.
- **6-level fill classification engine** via area-intersection geometry between detected stock and bin containers, resilient WMS submission through **Azure API Management** with local buffering and exponential retry, idempotent processing keyed by filename.

AEI (Accounting Engine Interface) - AI Document Processing & Validation Platform

Azure Logic Apps | Azure DI Studio | Content Understanding | GPT-4.1 | Angular 19 | Azure APIM | Azure DevOps

- End-to-end **accounting document platform** for Iberian retail clients (PT/ES): AI pipelines turn supplier invoices, delivery notes and product invoices into structured accounting data, and a web portal lets reviewers validate and correct each document before posting - processing time cut from **30 min to <2 min per document** at **200+ docs/day**.
- **Multi-model extraction**: a custom classifier routes invoices from ~110 **supplier VAT numbers** to supplier-specific cases combining custom **Azure DI Studio** models and **Content Understanding + GPT-4.1** pipelines, with a prebuilt + GPT-4.1 fallback for unknown suppliers, **95%+ classification** and **97%+ extraction accuracy**, and new suppliers onboard as a new case without redesign.
- Migrated the legacy ASP.NET MVC portal to an **Angular 19 SPA** calling **Azure API Management** directly with **OAuth2 Authorization Code + PKCE** and gateway JWT validation (no BFF), review screens for invoices, travel expenses with receipt preview, delivery notes and OCR documents, with inline editing and related-document tracking.
- **Fault-tolerant operations**: automatic error handling, incident tracking, multi-tier archiving for compliance and **per-invocation LLM cost tracking**, separate TEST/PROD environments and **Azure DevOps CI/CD** to Static Web Apps.

EDUCATION

ISTEC (Institute of Advanced Technologies PT) <i>Bachelor's Degree, Computer Engineering (EQF Level 6)</i>	Oct 2026 – Jul 2029
ISTEC (Institute of Advanced Technologies PT) <i>Associate's Degree, Networks and Computer Systems (EQF Level 5) - Grade: 16/20</i>	Oct 2023 – Jul 2026
IEFP - Instituto do Emprego e Formação Profissional <i>Certificate of Qualifications, Digital Commerce "Business Strategy" (EQF Level 4)</i>	Feb 2024 – Jul 2024
IPTA - Instituto Profissional de Tecnologias Avançadas <i>IT Equipment Management (EQF Level 4) - Grade: 16/20</i>	Sep 2020 – Jul 2023
Conservatório Regional de Música de Ferreirim <i>Euphonium Performance (5th Grade)</i>	Sep 2014 – Jun 2020

ADDITIONAL

Languages: Portuguese (Native) | English (Intermediate, B1)
Driving License: Category B (valid)
Availability: Upon request